RHEL System Administration – Scenario-Based Test

Scenario:
You are a Linux System Administrator for a college laboratory. A RHEL server is used for development and student data storage. During routine maintenance, you must configure the network, manage software, organize files, and prepare a separate filesystem for student data.

1. Network Configuration
Task: Configure 192.168.10.50/24, gateway 192.168.10.1 and DNS 8.8.8.8 using nmcli. Activate the connection.

2. Network Validation
Task: Verify the server's IP address, routing information and network connectivity using suitable commands.

3. Hostname
Task: Change localhost.localdomain to server01.lab.local and verify the new hostname.

4. Locate a File
Task: Locate lab.conf somewhere under /etc using an appropriate Linux command.

5. Compressed Archive
Task: Create project_backup.tar.gz from /home/student/project and verify its contents without extracting it.

6. Mount Filesystem
Task: For XFS partition /dev/sdb1, create /data, mount it there and verify that it is accessible.

7. Unmount Filesystem
Task: Verify whether /data is mounted and safely unmount the filesystem.

8. Symbolic Link
Task: Create /home/student/report.txt as a symbolic link to /data/project/report.txt and verify it.

9. Software Package
Task: Search for httpd, install it if required, and verify the installation and package information using RPM.

10. Software Updates
Task: Display enabled repositories, check available updates and update installed packages using Yum/DNF.